

Famine Early Warning Systems Network

# El Nino Forecast and its implications for the 2026/27 season and acute food insecurity in Angola and southern Africa

June 26  
Angola

# Agenda

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- El Nino Overview
- El Niño Southern Oscillation
- El Niño Characteristics
- El Niño Impacts
- El Niño Implications on Acute Food Insecurity
- Q&A

# Overview

## **The El Niño-Southern Oscillation**

*Naturally occurring, arising from tropical atmosphere-ocean relationship*

## **Forecast**

*El Niño likely to be moderate to strong by late 2026*

# Overview

## **The El Niño-Southern Oscillation**

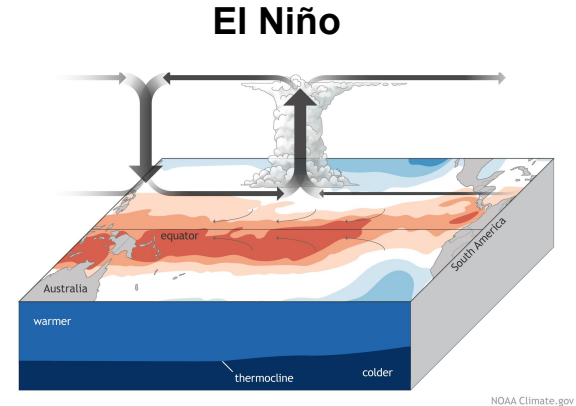
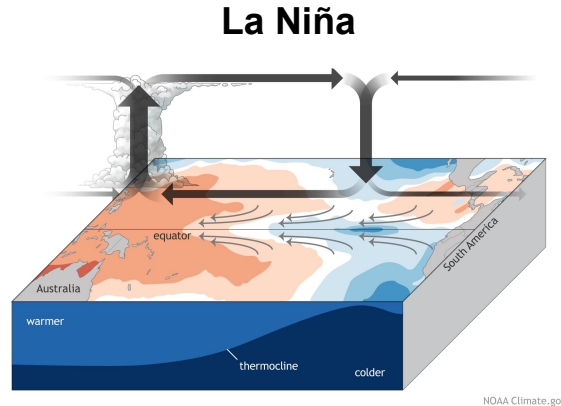
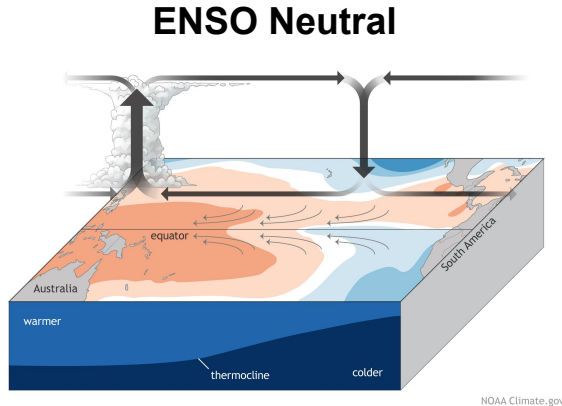
*Naturally occurring, arising from tropical atmosphere-ocean relationship*

## **Forecast**

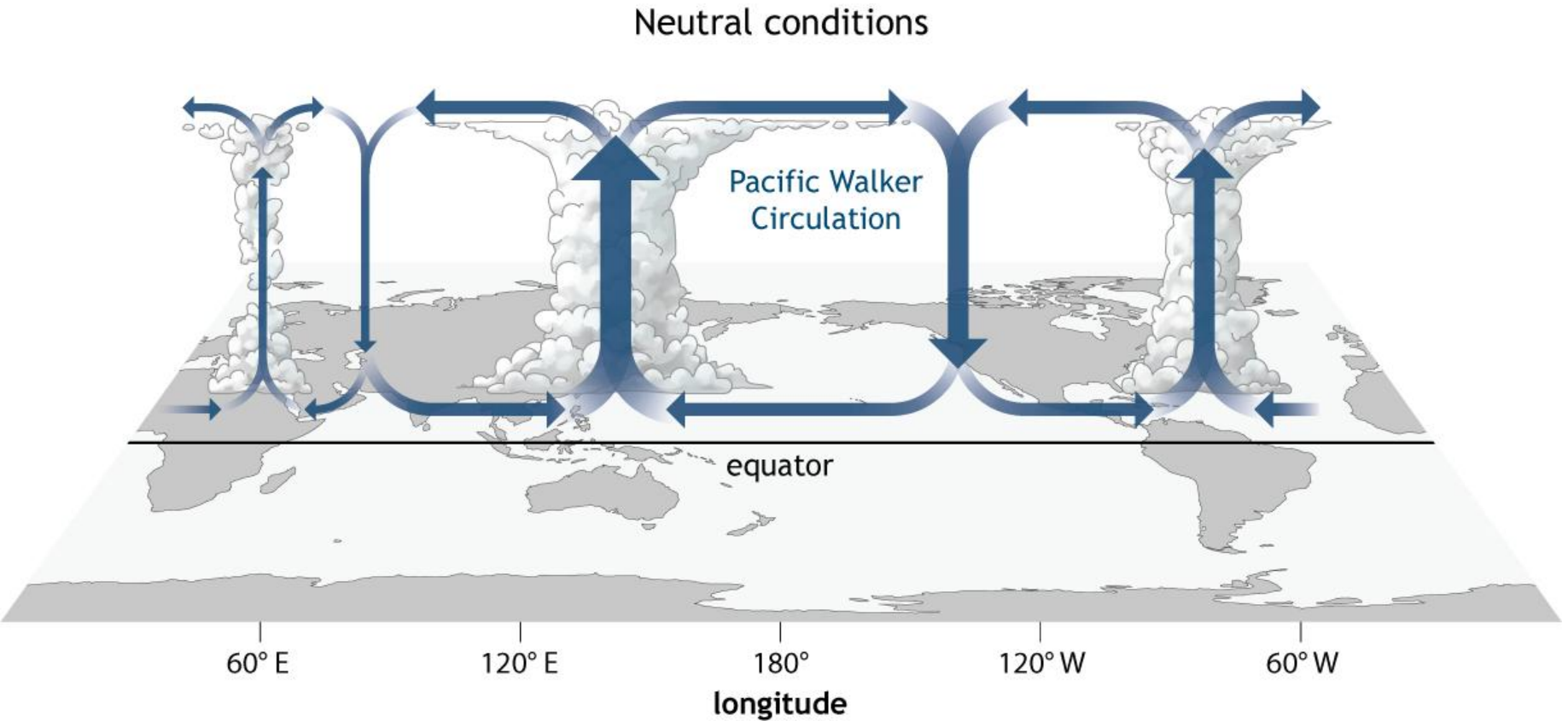
*El Niño likely to be moderate to strong by late 2026*

# The El Niño – Southern Oscillation (ENSO)

3 Phases:

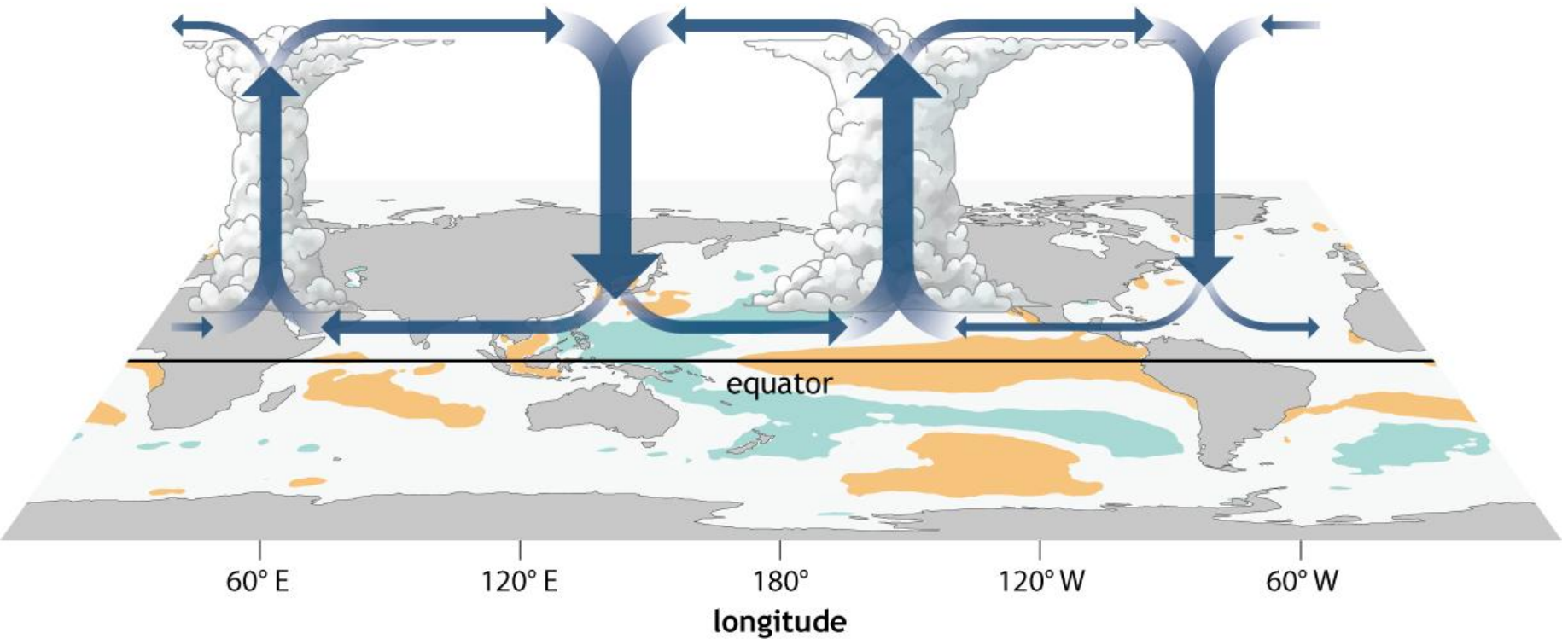


Changes in temperature and wind patterns in the Pacific Ocean have global effects



Changes in temperature and wind patterns in the Pacific Ocean have global effects

El Niño conditions

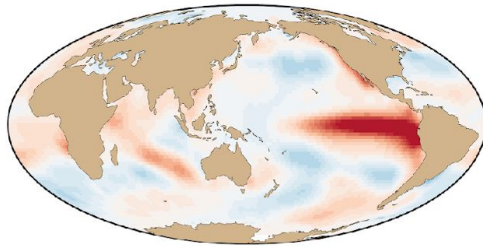


# No Two El Niño Events are the Same

Generally characterized by the strength of sea surface temperature departures from average (*intensity*) and the location of the departures (*flavor*)

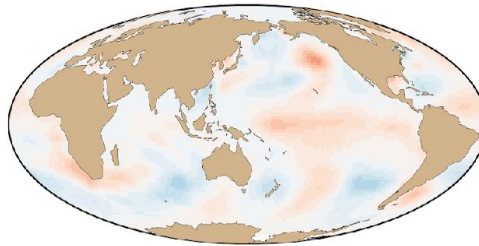
**Intensity:** Strong  
**Flavor:** Eastern Pacific

**1997-1998**



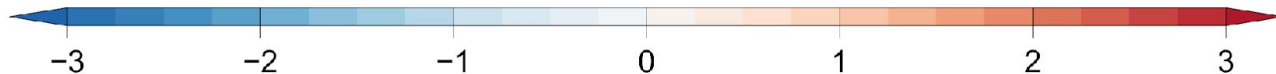
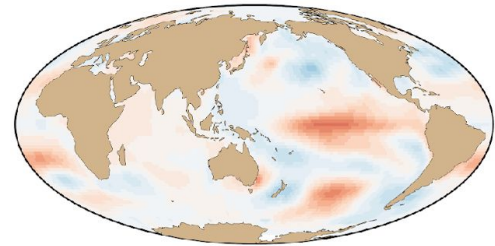
**Intensity:** Weak  
**Flavor:** Central Pacific

**2004-2005**



**Intensity:** Moderate  
**Flavor:** Central Pacific

**2009-2010**



Sea Surface Temperature Departure From 1951-2022 Average (°C)

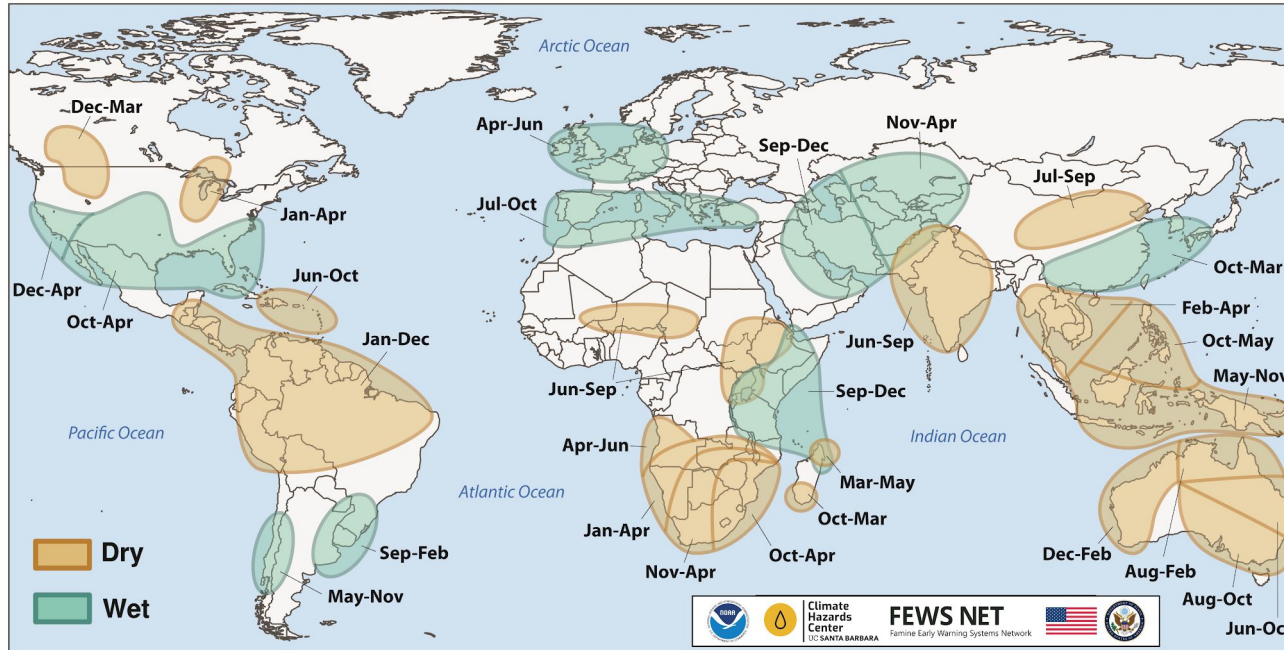
Source:  
NOAA  
Physical  
Sciences  
Laboratory





# Typical Global El Niño Impacts

## El Niño and Precipitation



# Overview

## **The El Niño-Southern Oscillation**

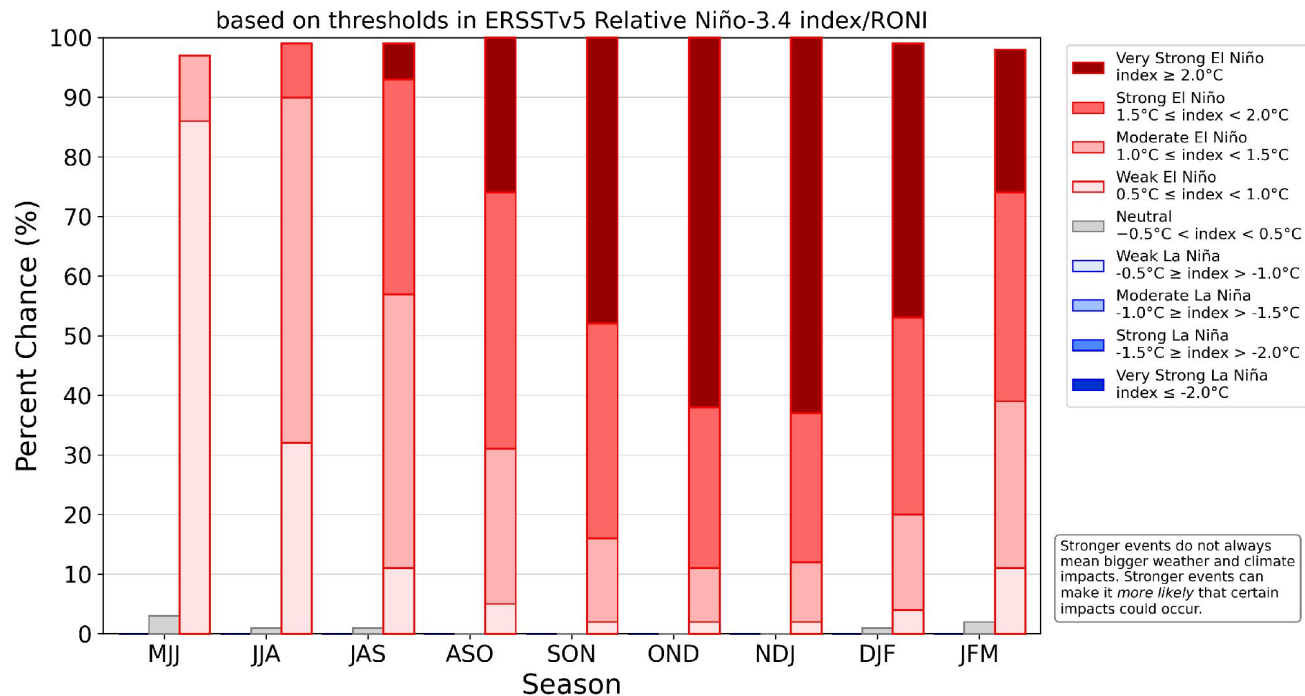
*Naturally occurring, arising from tropical atmosphere-ocean relationship*

## **Forecast**

*El Niño likely to be moderate to strong by late 2026*

# Latest El Nino predictions

## NOAA CPC ENSO Strength Probabilities (issued June 2026)

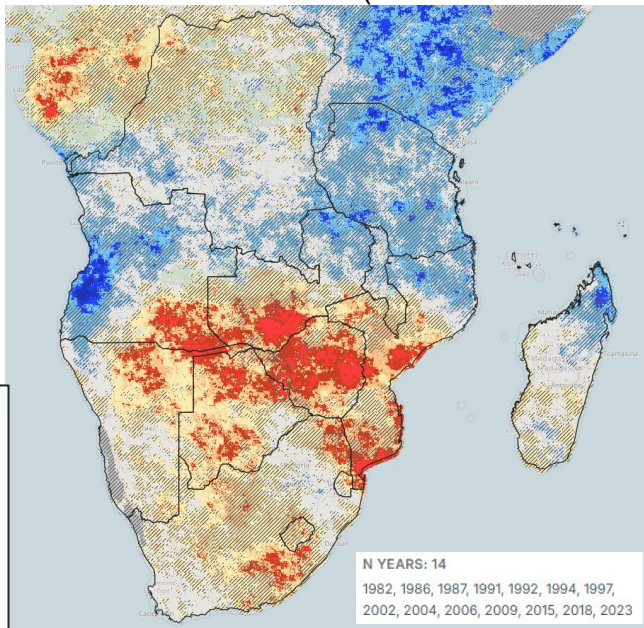


- El Nino forecast for the 2026/27 season, by strength

- **98% chance** of moderate El Nino or stronger in **OND and NDJ**
- **95% chance** of moderate El Nino or stronger in **DJF**

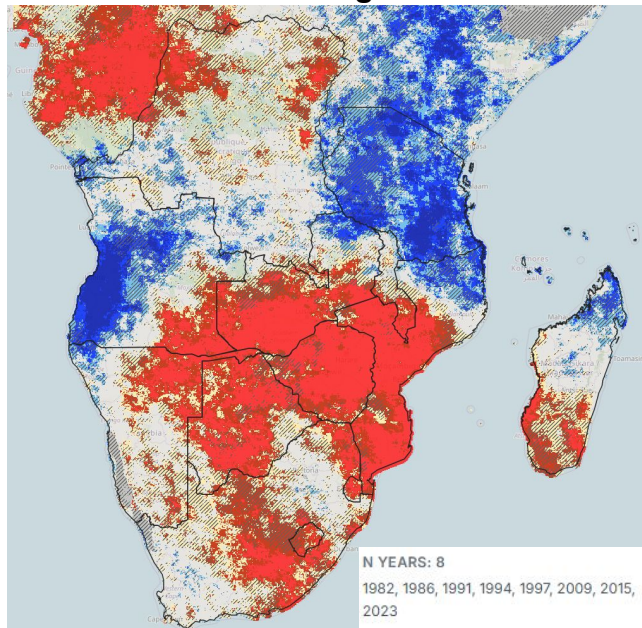
# Why El Niño Strength Matters

DJF rainfall frequencies (terciles),  
**All El Nino events (1981 to 2025)**



Source:  
Data: CHIRPS3, UCSB CHC  
Analysis: enso.ubramplab.org

DJF rainfall frequencies (terciles),  
**Moderate-to-Strong El Nino events**



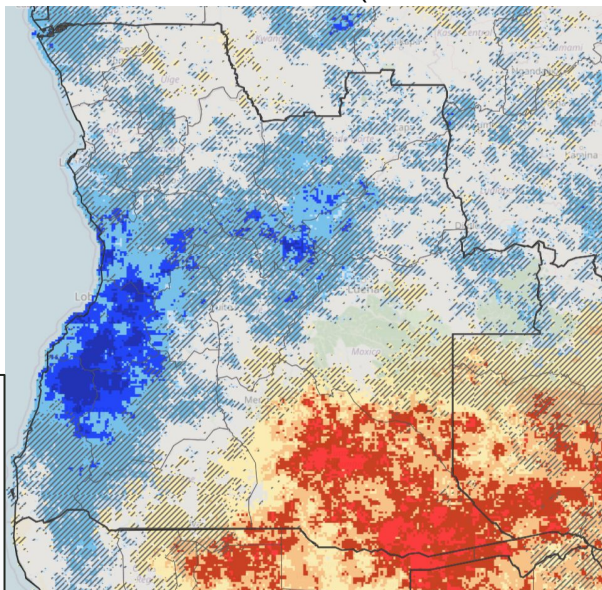
Source:  
Data: CHIRPS3, UCSB CHC  
Analysis: enso.ubramplab.org

- Probability of below normal rainfall in south and center increases significantly during moderate-to-strong El Niño, compared to all El Niño events.
- Similarly with above normal rainfall in north-east and north-west



# Why El Niño Strength Matters

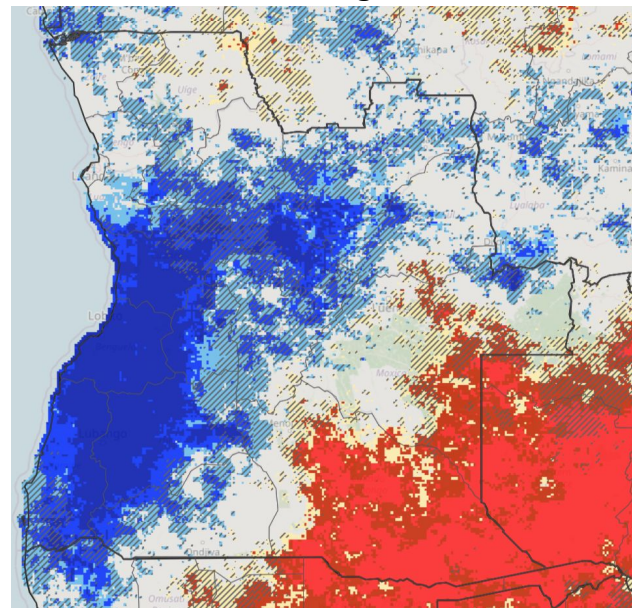
DJF rainfall frequencies (terciles),  
**All El Nino events (1981 to 2025)**



N YEARS: 14

Source:  
Data: CHIRPS3, UCSB CHC  
Analysis: [enso.ubramplab.org](http://enso.ubramplab.org)  
1982, 1986, 1987, 1991, 1992, 1994, 1997,  
2002, 2004, 2006, 2009, 2015, 2018, 2023

DJF rainfall frequencies (terciles),  
**Moderate-to-Strong El Nino events**

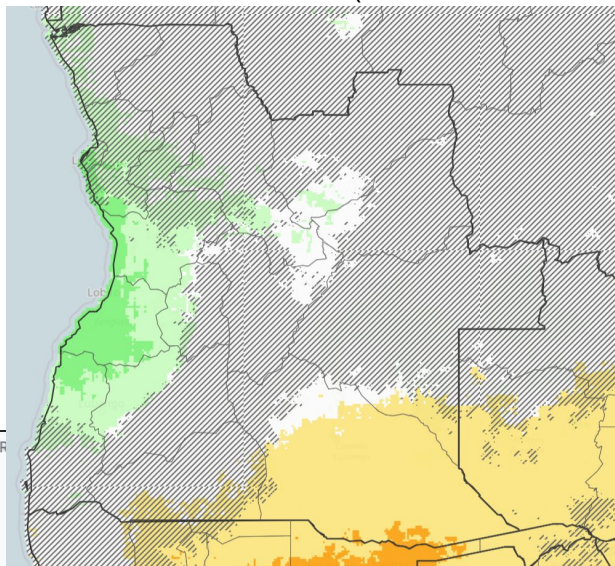


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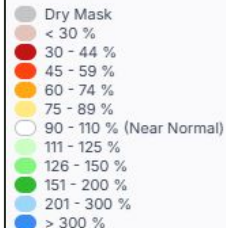
- Probability of below normal rainfall in south-east increases significantly during moderate-to-strong El Niño, compared to all El Niño events.
- Similarly with above normal rainfall in central and western Angola

# Why El Niño Strength Matters

DJF rainfall composites (pct of avg),  
**All El Nino events (1981 to 2025)**



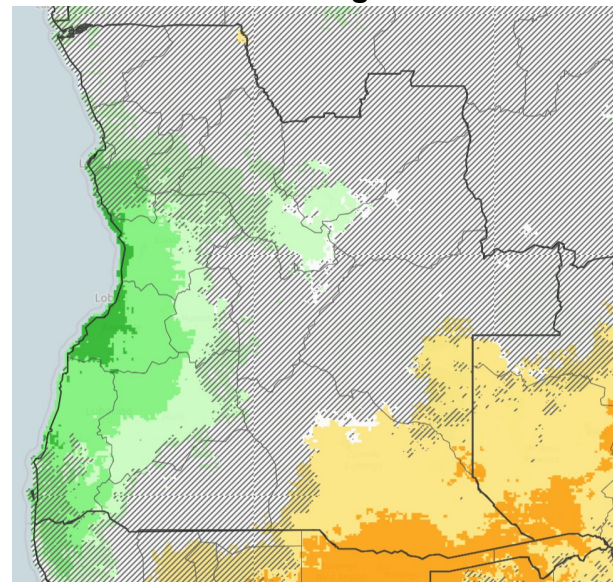
ENSO RAINFALL (% OF LONG-TERM AVERAGE)



N YEARS: 14  
1982, 1986, 1987, 1991, 1992, 1994, 1997,  
2002, 2004, 2006, 2009, 2015, 2018, 2023

Source:  
Data: CHIRPS3, UCSB CHC  
Analysis: enso.ubramplab.org

DJF rainfall composites (pct of avg),  
**Moderate-to-Strong El Nino events**



N YEARS: 8  
1982, 1986, 1991, 1994, 1997, 2009, 2015,  
2023

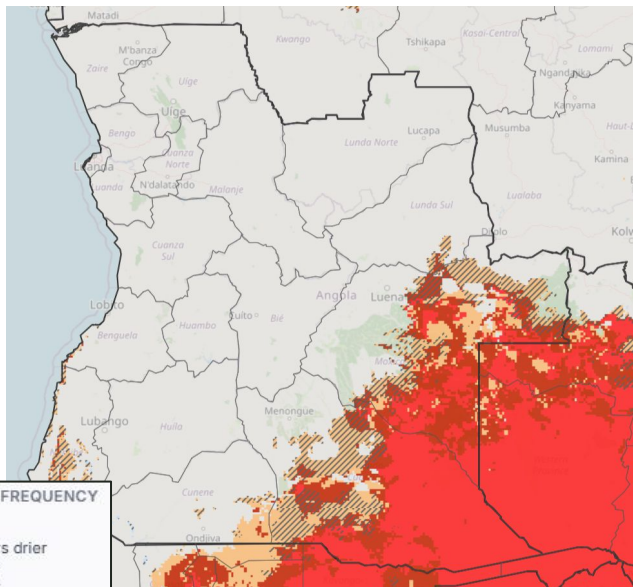
Source:  
Data: CHIRPS3, UCSB CHC  
Analysis: enso.ubramplab.org

*The average rainfall is lower in the south-east during moderate-to-strong El Niño, compared to all El Niño events*



# Other Indices show agreement for adverse outcome

**DJF Tmax > 35 C frequency, days with Moderate-to-Strong El Nino events**

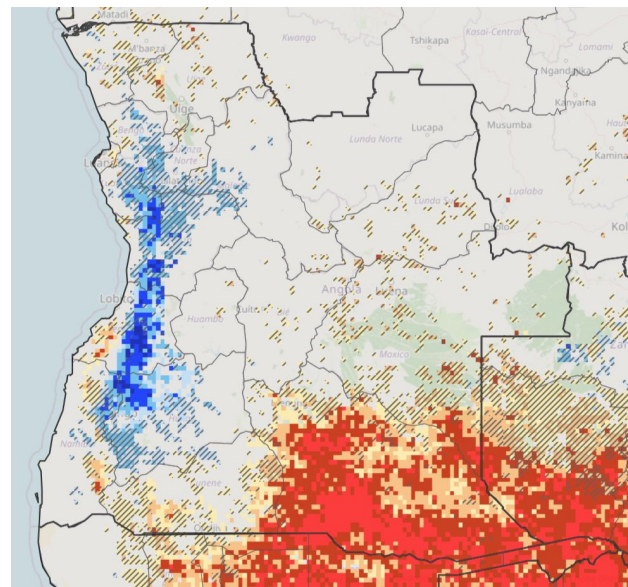


Source:  
Data: CHIRTS-ERA5, UCSB CHC  
Analysis: enso.ubramplab.org

N YEARS: 8

1982, 1986, 1991, 1994, 1997, 2009, 2015,  
2023

**WRSI frequency for below/above normal Moderate-to-Strong El Nino events**



Source:  
Data: WRSI

Analysis: enso.ubramplab.org

N YEARS: 8

1982, 1986, 1991, 1994, 1997, 2009, 2015,  
2023

## ENSO RAINFALL FREQUENCY ANOMALY

- >70% of years drier
- 60-70% drier
- 50-60% drier
- 40-50% drier
- Uncertain
- 40-50% wetter
- 50-60% wetter
- 60-70% wetter
- >70% wetter
- Dry mask

- Additional parameters provide similar outcomes under mod-to-strong El Nino:
- Extreme temperature, WRSI, dry spells, evapotranspiration, SPEI...

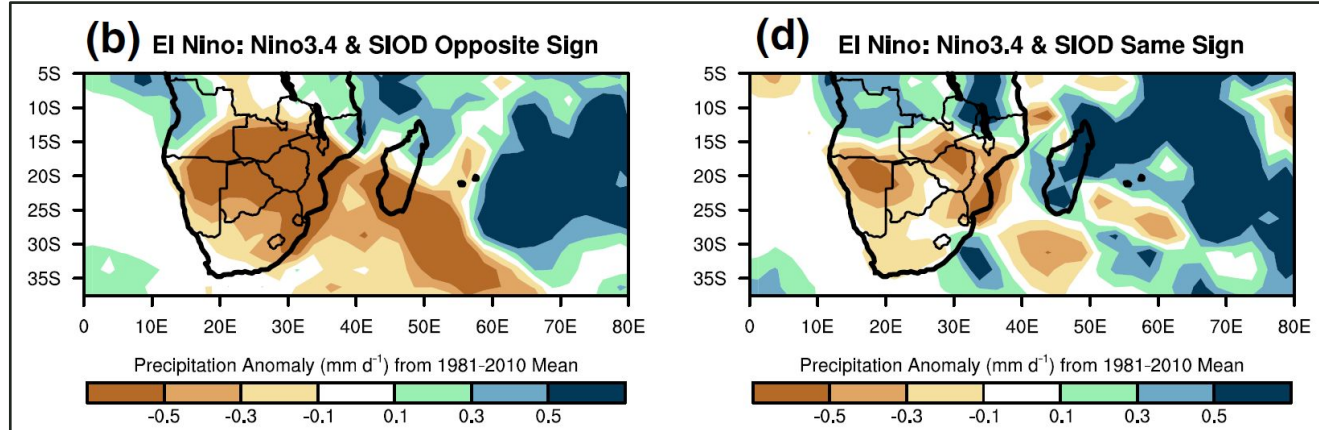
**FEWS NET**

Famine Early Warning Systems Network



# Regional Influences

- Regional climate drivers modulate ENSO impacts in southern Africa
  - Subtropical Indian Ocean Dipole (SIOD), Angola Low
- Regional drivers have shorter lead-time predictability than ENSO, improve forecast updates



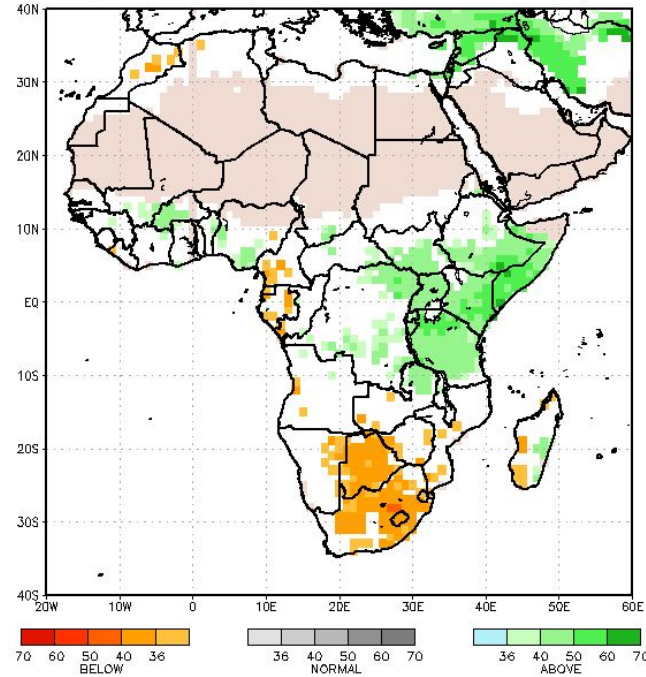
Source: Hoell et al., 2017

Example modulation: negative SIOD can moderate El Niño impacts

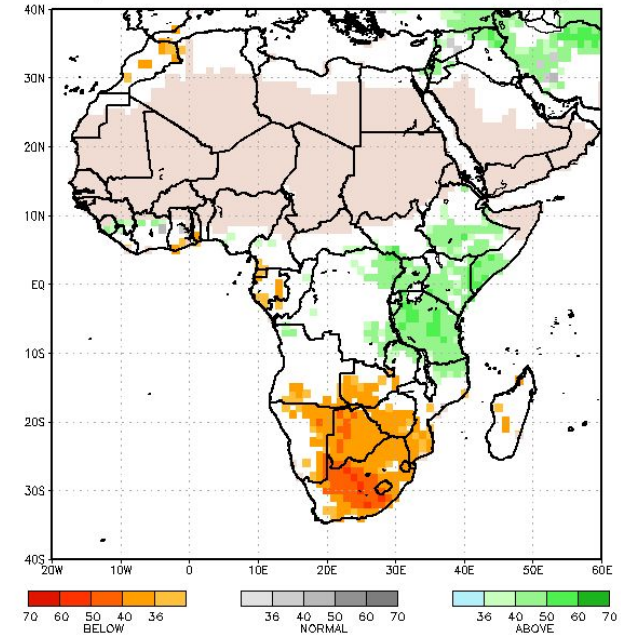


# NMME forecast for OND & NDJ reflect typical El Nino impacts

NMME Precip Prob. JunIC Oct2026-Dec2026 Fcst Sand color: Oct-Dec DryClim Mask



NMME Precip Prob. JunIC Nov2026-Jan2027 Fcst Sand color: Nov-Jan DryClim Mask

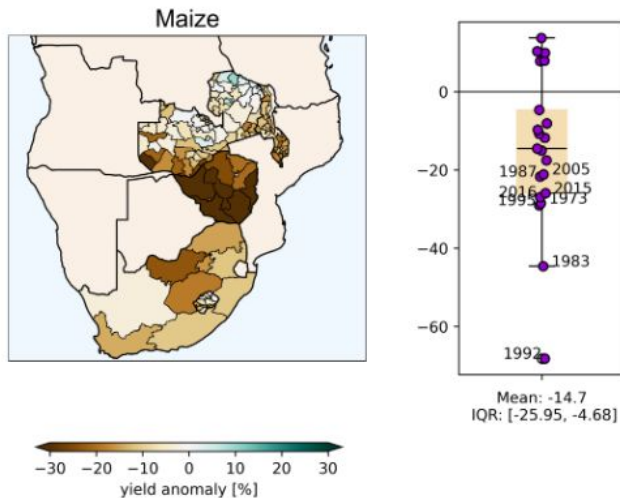


# Maize yields are often below normal in Southern Africa

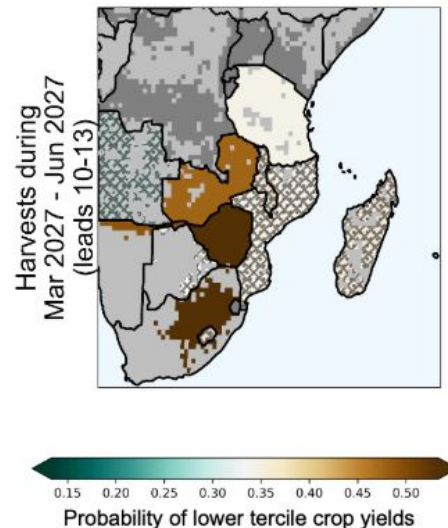
## Analog year analysis and ENSO-based forecast

- During El Niño years, maize yields tend to be below-normal, with regional total yields being -15% below normal on average
- El Niño-based maize yield forecast in Southern Africa → 2027 harvest likely to be below normal, particularly in Zimbabwe, South Africa, Namibia
- Hatched areas → no forecast skill at this lead; data

### Maize yields during El Niño years



### Maize yield forecast



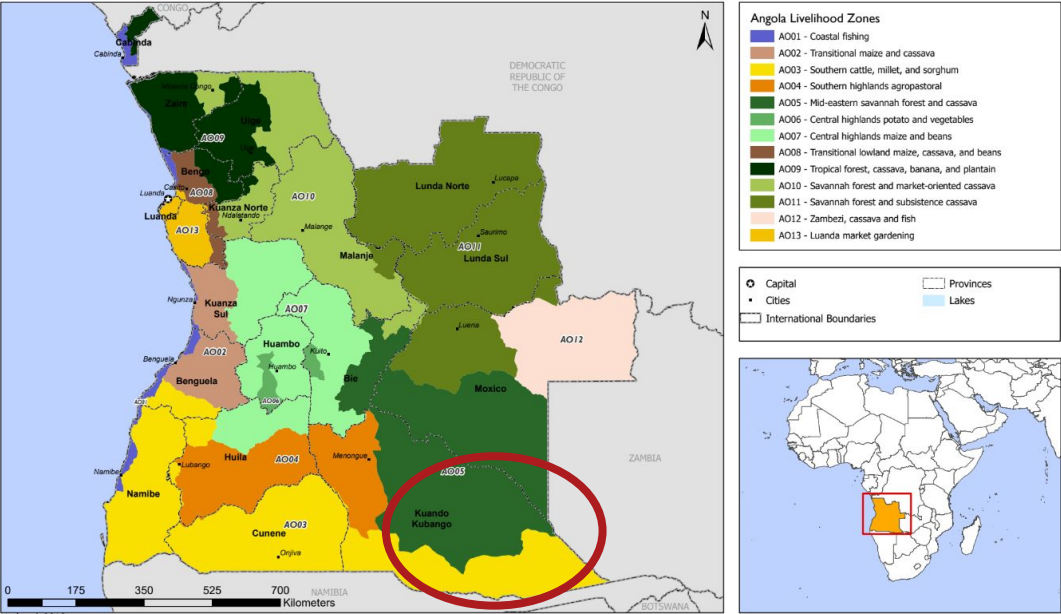
# Summary

- High confidence (95%+ chance) of moderate to strong El Niño in OND-DJF
- El Niño results in a high probability of
  - below average rainfall in south-eastern Angola
  - Above average rainfall in western and central Angola
- Moderate-to-strong El Niño increases probability of high temperatures, low rainfall, dry spells
- Not every El Niño has the same impacts – impacts are probabilistic
- Some central and northern parts of Angola have lower certainty of El Niño -induced rainfall outcomes
- Regional climate drivers can modulate El Niño impacts - continual monitoring and updated forecasts required throughout
- Availability of historical yield data poses challenges for modelling harvest outcomes

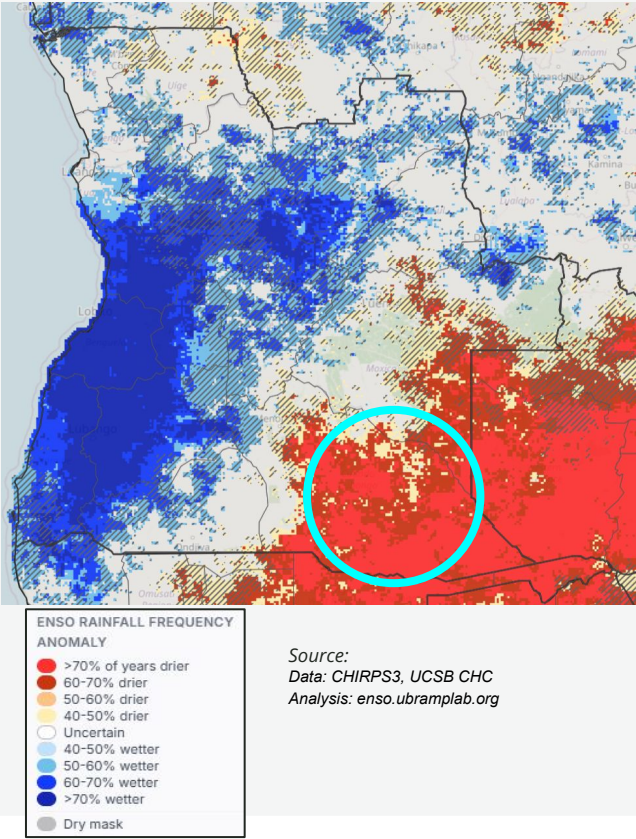
# Implications of the El Niño on Acute Food Insecurity in Angola



# Angola Livelihood Zones likely to be most affected by the El Nino



Source: FEWS NET

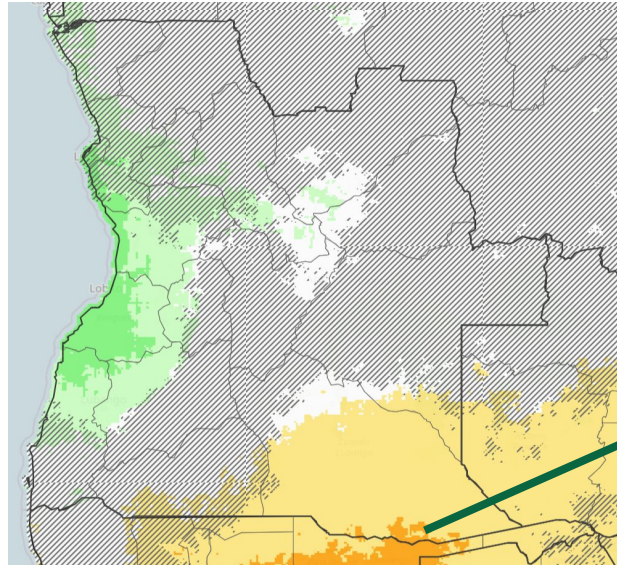


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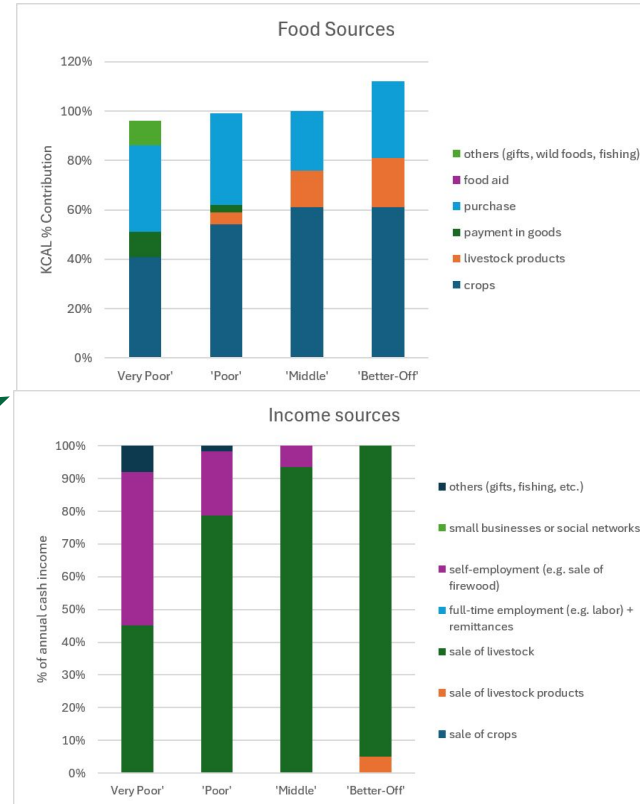
# Household exposure to the shock: Livelihood Zone AO03

DJF rainfall composites (pct of avg),  
All El Nino events (1981 to 2025)



Source:  
Data: CHIRPS3, UCSB CHC  
Analysis: enso.ubramplab.org

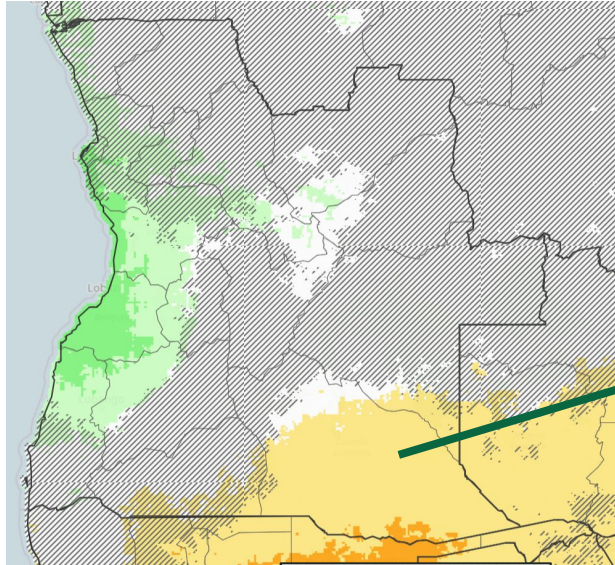
## Livelihood Zone AO03: Southern cattle millet and sorghum



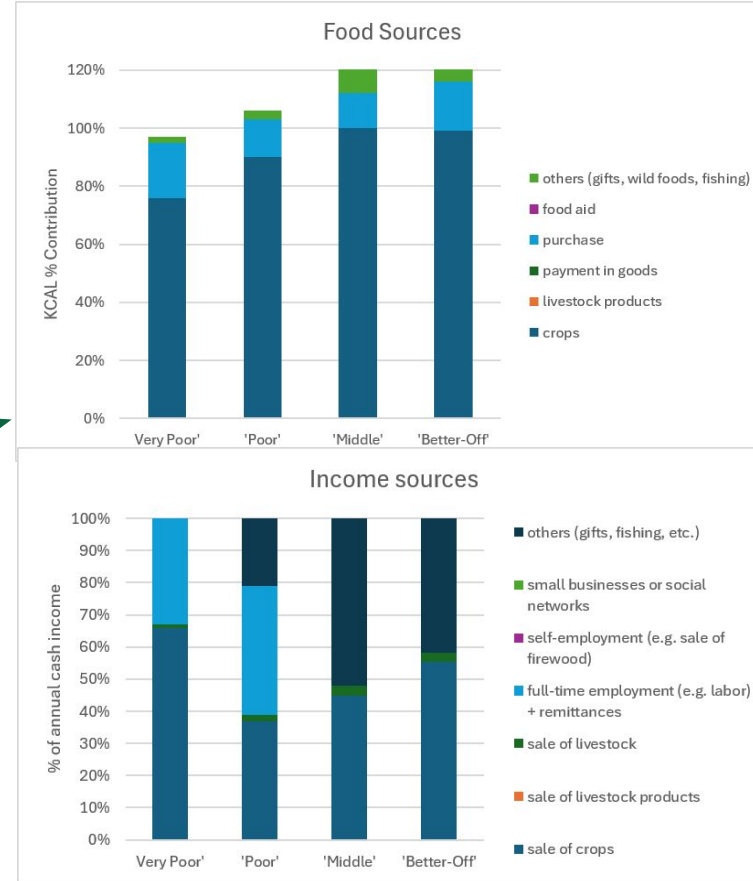
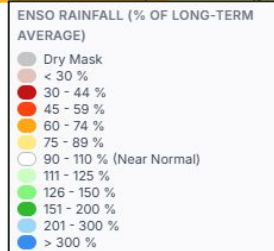
# Household exposure to the shock: Livelihood Zone AO05

Livelihood AO05: Mid-eastern savanna forest and cassava

DJF rainfall composites (pct of avg),  
All El Nino events (1981 to 2025)



Source:  
Data: CHIRPS3, UCSB CHC  
Analysis: enso.ubramplab.org



# Southern Africa timing of peak needs

Prepare for likely increase in food assistance needs in 2027/28, peaking in **January-March 2028**.



Current  
annual  
harvest &  
post-harvest  
period

Increased likelihood of  
▼ **below-average rainfall**  
during main crop growing  
season

Increased  
likelihood of  
▼ **below-average  
cereal harvests**  
in 2027

▼ **Household food  
consumption  
deteriorates as food  
prices peak**

Next  
annual  
harvest  
period